

The University of Lahore, CS & IT Department

Object Oriented Programming

Lab Assignment # 02

Part - 2: Hotel Management System

Start Date: 01/05/2025

Total Marks: 20

Due Date: 10/05/2025

Program: BSCS

Instructions

1. Understanding the problems is part of the assignment. So, no query, please.
 2. You will get zero marks if found any type of plagiarism or AI generated code.
 3. No submission after due date.
 4. Submit the .cpp file containing your code.
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Q1. Hotel Structure and Room Management using Linked List

1. Objective

You are required to implement a hotel room and floor management system using the Linked List data structure. This system models the structure of the Galaxy Hotel where every floor contains rooms, and each room has attributes such as type and status.

2. Scenario: Galaxy Hotel – Floor and Room Organizer

The Galaxy Hotel has expanded its operations and introduced a modular structure for room and floor data management. Instead of using static arrays or databases, they want a dynamic system using Linked Lists, where:

- Each floor is a node in a singly linked list.
- Each floor node contains a linked list of rooms.
- This allows dynamic expansion, removal, and traversal of floors and rooms at runtime.

You are the system architect assigned to build this flexible floor-room layout manager.

3. System Overview

Each room has:

- Room ID (unique per hotel, e.g., 201 for floor 2 room 1)
- Room Type: Single, Double, Suite
- Status:
 - Ready (available to book)
 - Booked (scheduled)
 - Occupied (currently in use)
 - Unavailable (under maintenance)

4. Required Functionalities

4.1. Create Hotel Structure

- Take input for:
 - Number of floors (F)
 - Number of rooms per floor (N)
- Use linked lists to dynamically create F floor nodes.
- Each floor node should contain a list of N room nodes.

4.2. Assign Room Types

- Let the user assign a room type for each room.

4.3. Update Room Status

- Allow updating the status of any room by entering the room ID.

4.4. Retrieve Room Information

- Display information for a specific room.
- View all rooms on a specific floor.

4.5. Summary and Analysis

- Count of:
 - Total rooms
 - Rooms by type (Single, Double, Suite)
 - Rooms by status (Booked, Ready, Occupied, Unavailable)
- List all available (Ready) rooms by type.

4.6. Remove a floor and all its rooms.

4.7. Add a new floor dynamically at runtime.

4.8. Search rooms by type across all floors.

4.9. Display hotel layout in floor-wise tabular view..

5. Program Requirements

- Use singly or doubly linked lists only — do not use arrays or STL/Java collections.
- Use classes/structs to represent:
 - Floor
 - Room
- Maintain clean code structure and user-friendly interface.